Data-driven adjoint-based algorithms for estimation, learning, and control

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Idea and main result

Data-driven control

- highly popular
- this IFAC: Data-Driven Control Theory I, II, III, ...
- many approaches based on Willems' fundamental lemma
  - e.g., DeePC
  - non-parametric representation implicitly encoded in the data
  - see `WeC06.3` for connections to subspace predictive control
  - also used in ILC(Jiang & Chu2022) and repetitive control(Dinkla et al.2024)

Today: data-driven adjoint-based algorithms

- do not implicitly estimate a model
- only very mild assumptions (LTI)
- adjoints are at the heart of the field
  - role in optimal control, minimum energy control, and optimal ILC
  - key in frequency-domain ILC (`filtfilt`)
  - mathematical foundation for duality controllability and observability

Idea and main result

Some well-known algorithm where we use models

1. Iterative Learning Control (ILC) $f_{j+1} = f_j + Le_j$,
   with L typicallly based on a model
2. Uncertainty estimation for robust control $y = \Delta f$,
   where a model $\hat{\Delta}$ is of Δ is made, and $\|\hat{\Delta}\|_\infty$ is computed

Today: unified framework for data-driven counterpart

- adjoint-based for estimation, learning, and control
  - optimal ILC
  - system norms
  - system properties
  - ...
- many extensions, multivariable, resetting, etc.

Two algorithms in more detail

ILC

criterion:

$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_{W_e} + \|f_{j+1}\|_{W_f}$$

is minimized by iteration:

$$f_{j+1} = (I - \epsilon_j W_f) f_j + \epsilon_j J^T W_e e_j$$

$\mathcal{H}_\infty$

estimator:

$$\beta_j = \sqrt{\frac{z_j^T f_j}{f_j^T f_j}}$$

converges to $\mathcal{H}_\infty$ norm via iteration:

$$f_{j+1} = \epsilon_j J^T J f_j$$

Key observation: involves J^T

On J^T

- $$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{N-1} \end{bmatrix}}_{y} = \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_{J} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{bmatrix}}_{u}$$
- traditional model-based approaches: make model $\hat{J}$ of J, then take $\hat{J}^T$
- the key idea

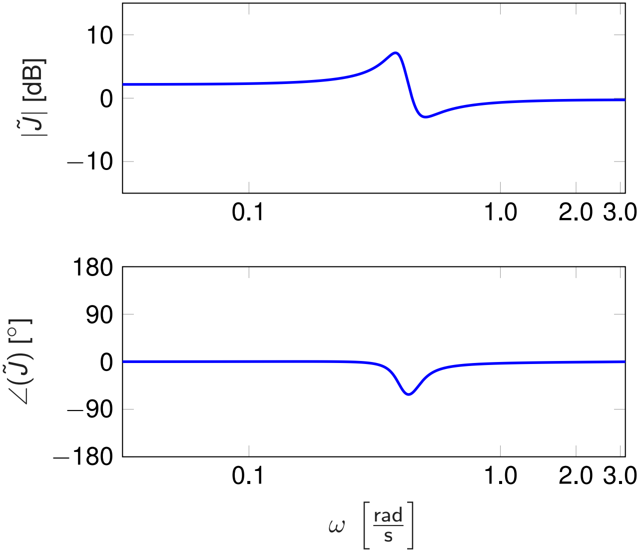
$$\underbrace{\begin{bmatrix} h_0 & h_1 & \cdots & h_{N-1} \\ 0 & h_0 & \ddots & h_{N-2} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & h_0 \end{bmatrix}}_{J^T} = \underbrace{\begin{bmatrix} 0 & \cdots & 0 & 1 \\ \vdots & \cdot^{\cdot^{\cdot}} & \cdot^{\cdot^{\cdot}} & 0 \\ 0 & \cdot^{\cdot^{\cdot}} & \cdot^{\cdot^{\cdot}} & \vdots \\ 1 & 0 & \cdots & 0 \end{bmatrix}}_{\mathcal{T}} \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_{J} \underbrace{\begin{bmatrix} 0 & \cdots & 0 & 1 \\ \vdots & \cdot^{\cdot^{\cdot}} & \cdot^{\cdot^{\cdot}} & 0 \\ 0 & \cdot^{\cdot^{\cdot}} & \cdot^{\cdot^{\cdot}} & \vdots \\ 1 & 0 & \cdots & 0 \end{bmatrix}}_{\mathcal{T}}$$

- evaluate J^T by: time reversal $\mathcal{T} \Rightarrow$ filter through $J \Rightarrow$ time reversal $\mathcal{T}$
- J^T is the adjoint operator (recall `filtfilt` in ILC)

Application to an example system

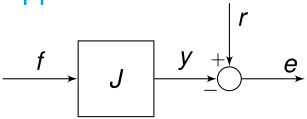
Example system

- transfer function $\tilde{J}$
- convolution matrix J with $N = 100$
- benchmark system (see extended paper for many use cases)



Application to an example system

Application: ILC

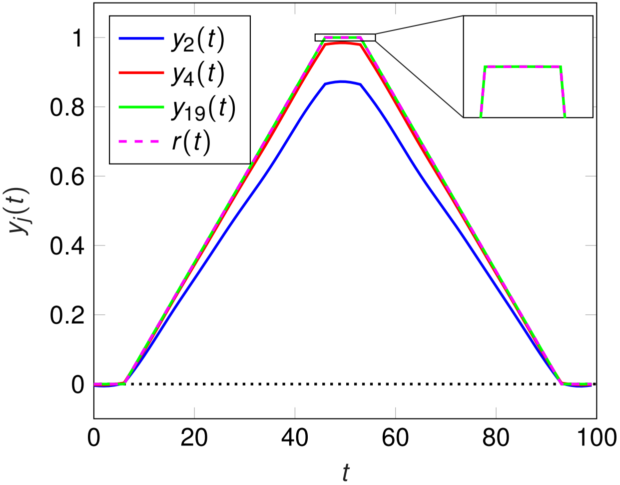


- criterion

$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_2$$

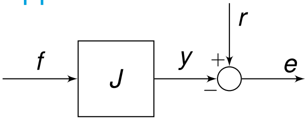
- minimized by iteration

$$f_{j+1} = f_j + \epsilon_j J^T W_e e_j$$



Application to an example system

Application: ILC



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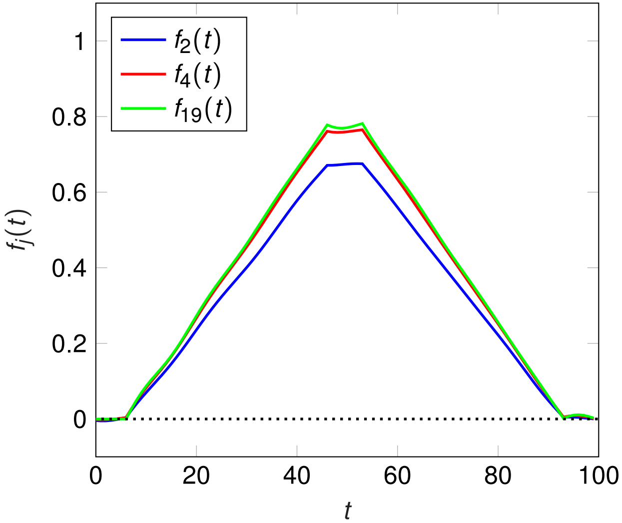
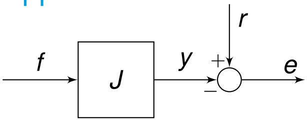


Application to an example system

Application: ILC



- criterion

$$\mathcal{J}(f_{j+1}) = \|e_{j+1}\|_2$$

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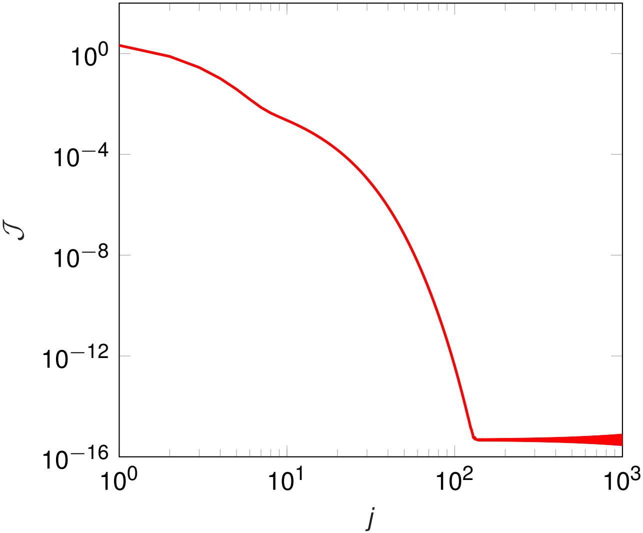
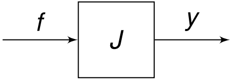


Application to an example system

Application: $\mathcal{H}_\infty$

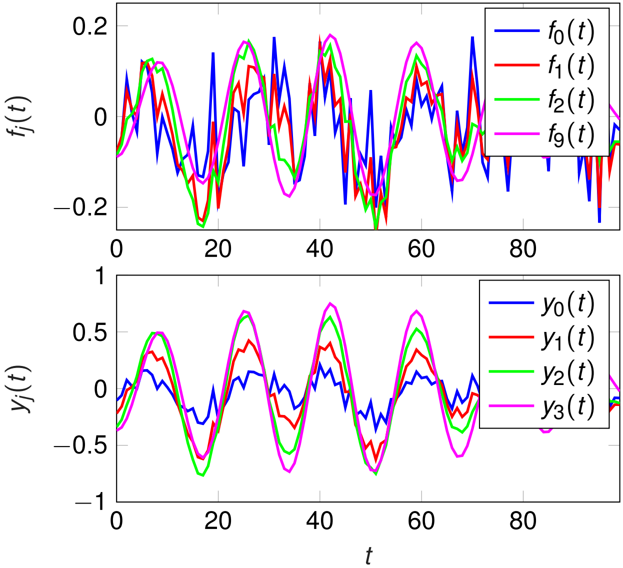


- estimator

$$\beta_j = \sqrt{\frac{z_j^T f_j}{f_j^T f_j}}$$

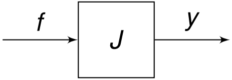
- with iteration:

$$f_{j+1} = \epsilon_j J^T J f_j$$



Application to an example system

Application: $\mathcal{H}_\infty$



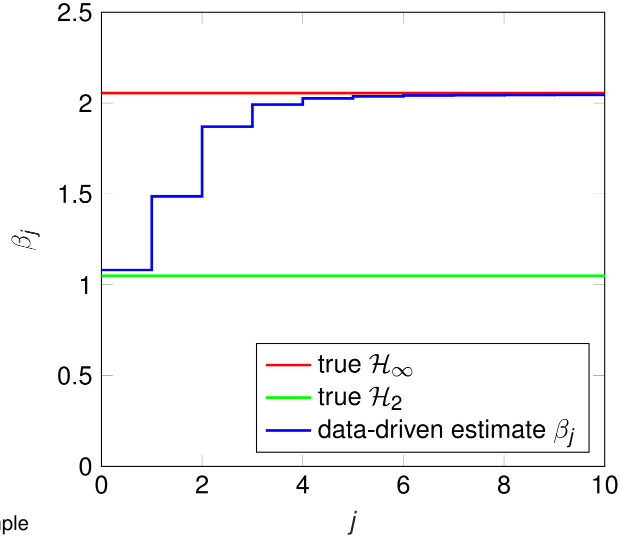
- estimator

$$\beta_j = \sqrt{\frac{z_j^T f_j}{f_j^T f_j}}$$

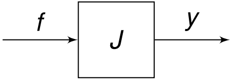
- with iteration:

$$f_{j+1} = \epsilon_j J^T J f_j$$

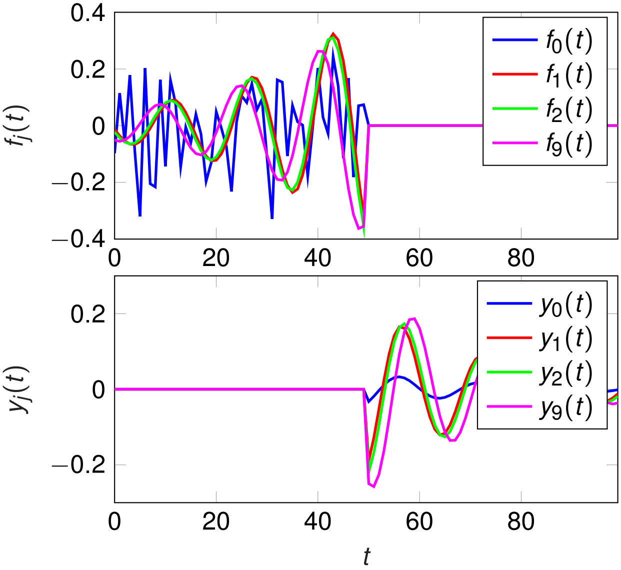
- Observations
  - $\beta_j \to \|J\|_\infty$, for $N \to \infty$
  - $\beta_0 \approx \|J\|_2$ (proof available)
  - we can do much more - next example



Application: Hankel norm

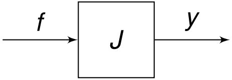


- accurate data-driven Hankel norm estimation
- interpretation Hankel norm: induced norm of past inputs to future outputs

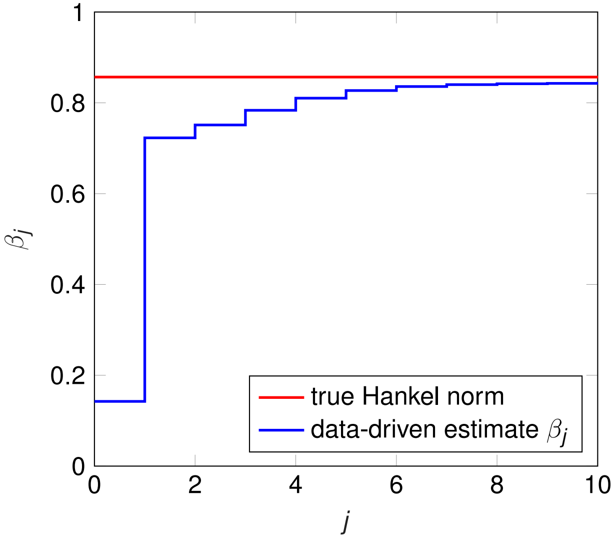


Application to an example system

Application: Hankel norm



- accurate data-driven Hankel norm estimation
- interpretation Hankel norm: induced norm of past inputs to future outputs
- many more system properties (see forthcoming paper)



Data-driven estimation: behind the algorithm

How to estimate norms of J without modeling?

$$f \longrightarrow \boxed{J} \longrightarrow y$$

- infinite time $\tilde{J}$: $\|\tilde{J}\|_\infty = \sup\limits_{f \in \ell_2} \dfrac{\|y\|_2}{\|f\|_2}$
- finite time J:
  - experiment length N: $$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}_{=y} = \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \cdots & 0 \\ \vdots & & \ddots & \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_{=J} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix}}_{=f}$$
  - then: $\sup\limits_{f \in \ell_2} \dfrac{\|y\|_2}{\|f\|_2} = \sup\limits_{f \in \ell_2} \sqrt{\dfrac{y^T y}{f^T f}} = \sup\limits_{f \in \ell_2} \sqrt{\dfrac{f^T J^T J f}{f^T f}}$
  - maximal if f corresponds to the eigenvector $\bar{f}$ of $\lambda^{\max}(J^T J)$:

$$\sup_{f \in \ell_2} \frac{\|y\|_2}{\|f\|_2} = \sqrt{\frac{\bar{f}^T J^T J \bar{f}}{\bar{f}^T \bar{f}}} = \sqrt{\lambda^{\max}(J^T J)}$$

$\Rightarrow$ connection $\|G\|_\infty$ and $\lambda^{\max}(J^T J)$

Data-driven estimation: behind the algorithm

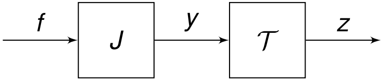
Analysis of $J^T J$

- as before: exploit Toeplitz structure

$$J^T = \begin{bmatrix} h_0 & h_1 & \cdots & h_{N-1} \\ 0 & h_0 & \cdots & h_{N-2} \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & h_0 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{=\mathcal{T} \text{ (time reversal)}} \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 \\ h_1 & h_0 & \cdots & 0 \\ \vdots & & \ddots & \\ h_{N-1} & h_{N-2} & \cdots & h_0 \end{bmatrix}}_{=J} \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{=\mathcal{T}}$$

- hence: $J^T J = (\mathcal{T}J)(\mathcal{T}J)$
- Hankel structure: $\mathcal{T}J = (\mathcal{T}J)^T \Rightarrow$ $\boxed{|\lambda^{\max}(\mathcal{T}J)| = \sqrt{\lambda^{\max}(J^T J)}}$
- thus: $|\lambda^{\max}(\mathcal{T}J)| \to \|G\|_\infty$ for $N \to \infty$
- remark: $\lambda^{\max}(J) = h_0$

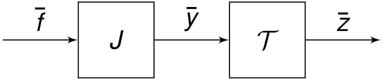
Resulting matrix $\mathcal{T}J$:



Data-driven estimation: behind the algorithm

We have: $|\lambda^{\max}(\mathcal{T}J)| \to \|\tilde{J}\|_\infty$ for $N \to \infty$

- Given $\bar{f}$ of $\lambda^{\max}(\mathcal{T}J)$, with $\|\bar{f}\|_2 = 1$. Evaluate:



$$\Rightarrow \quad \lambda^{\max}(\mathcal{T}J) = \bar{f}^T\bar{z}$$

- Key problem: how to obtain $\bar{f}$?

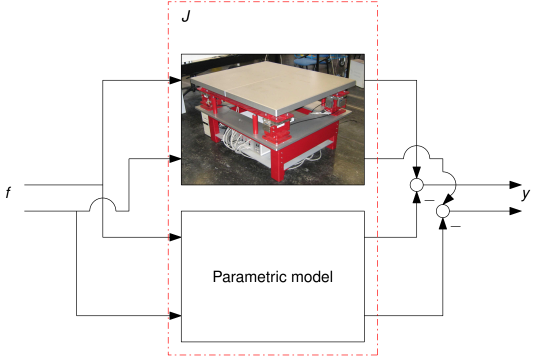
  $\Rightarrow$ power iteration (linear algebra): $x_{j+1} = \frac{Ax_j}{\|Ax_j\|}$, then $\lambda_k = x_j^T A x_j$ converges to $\lambda^{\max}(A)$

  Result: $f_j \to \bar{f}$ for $j \to \infty$ (global convergence)

Iteratively learning $\mathcal{H}_\infty$-norm (SISO)

- replace matrix ('model') J by evaluation on the true system
- then, $\beta_j = |f_j^T z_j| \to \|G\|_\infty$ for $j \to \infty$

Data-driven estimation: experimental case study



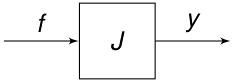
Motivation: robust control

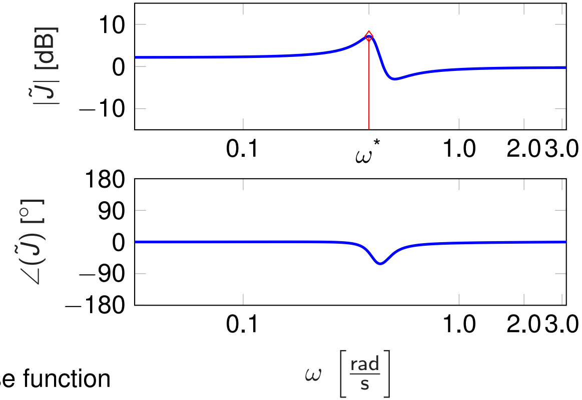
- model uncertainty bound γ
- if $\|J\|_\infty \leq \gamma$: guarantees!
- if $\|J\|_\infty > \gamma$: no guarantees
- if $\|J\|_\infty \ll \gamma$: conservative

Need accurate bound $\|J\|_\infty$

Frequency domain robust control

- J SISO, LTI, discrete time stable
  frequency domain interpretation:
  $\|\tilde{J}\|_\infty = \sup_{\omega} |\tilde{J}(e^{j\omega})| = |\tilde{J}(e^{j\omega^*})|$
- J MIMO, LTI, discrete time stable
  frequency domain interpretation:
  $\|\tilde{J}\|_\infty = \sup_{\omega} \bar{\sigma}\left(\tilde{J}(e^{j\omega})\right)$
  data-driven approach in (Oomen et al.2014)





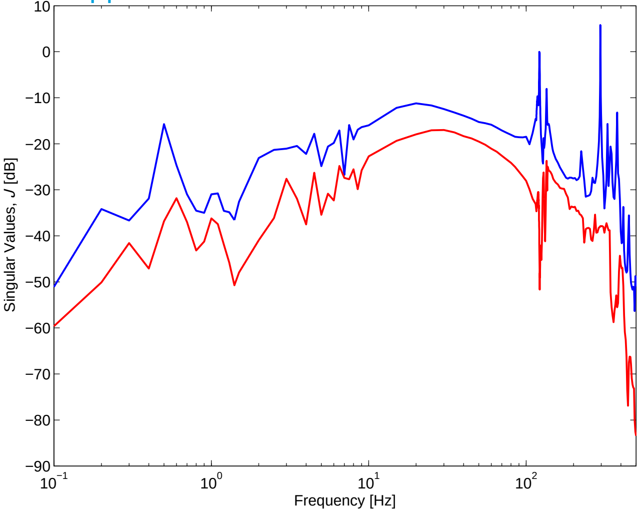
Classical model-based approaches

- identify parametric model $\hat{J}(\theta)$
  $\Rightarrow$ compute $\|J(\theta)\|_\infty \Rightarrow$ model quality?
- identify non-parametric frequency response function
  $\Rightarrow$ intergrid error

Data-driven estimation: experimental case study

Traditional frequency response function approach

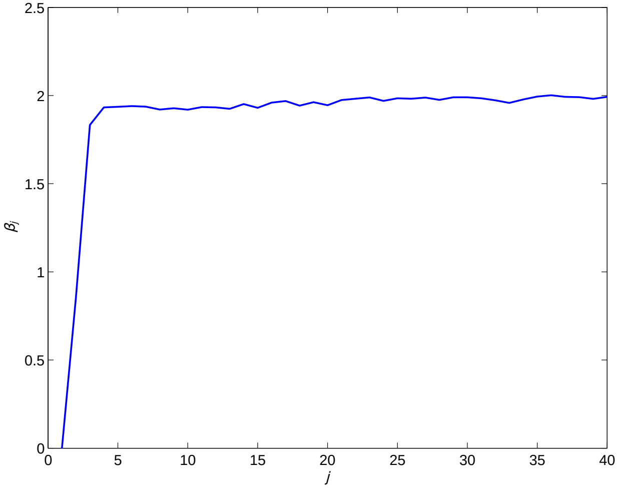
- Nonparametric identification of J
- long measurement
- $\|J\|_{\infty,FRF} = 1.95$
- $\omega^* \approx 296$ Hz



Data-driven estimation: experimental case study

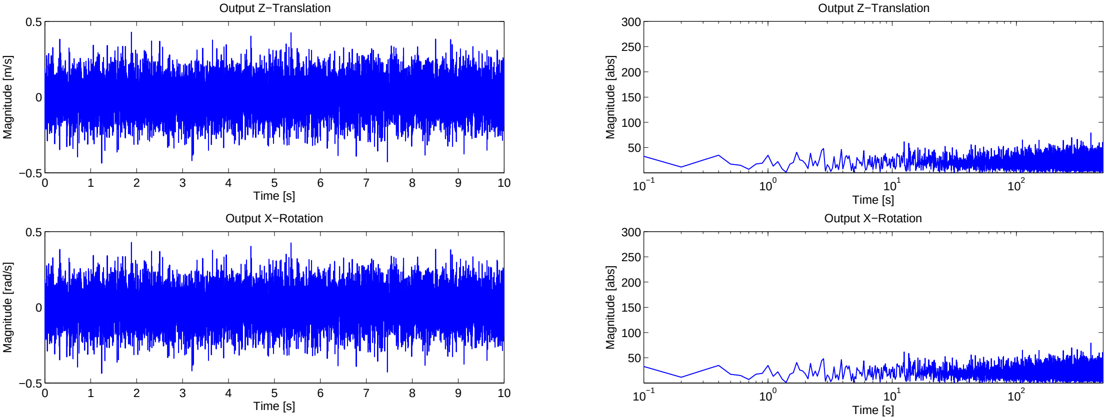
Data-driven algorithm

- $\|J\|_{\infty}^{\text{data}-\text{driven}} = 1.997$
- $(\|J\|_{\infty}^{FRF} = 1.95)$



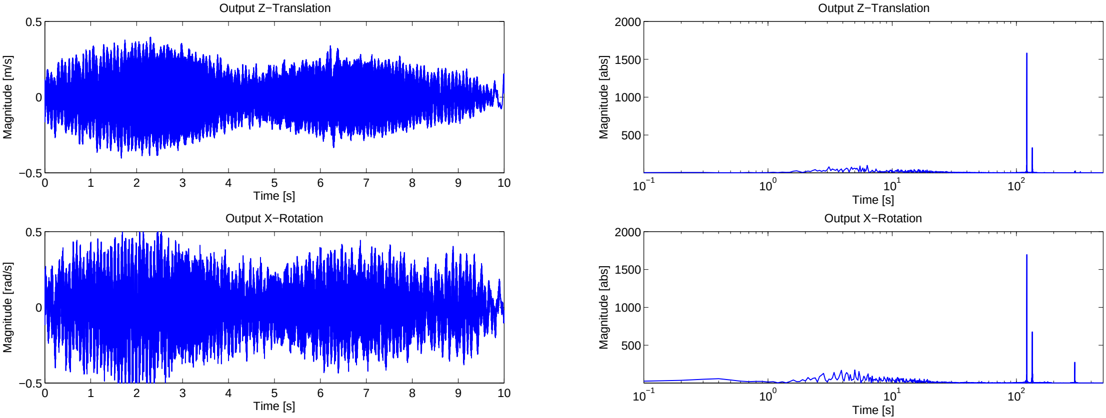
Data-driven estimation: experimental case study

- Initial input: white noise



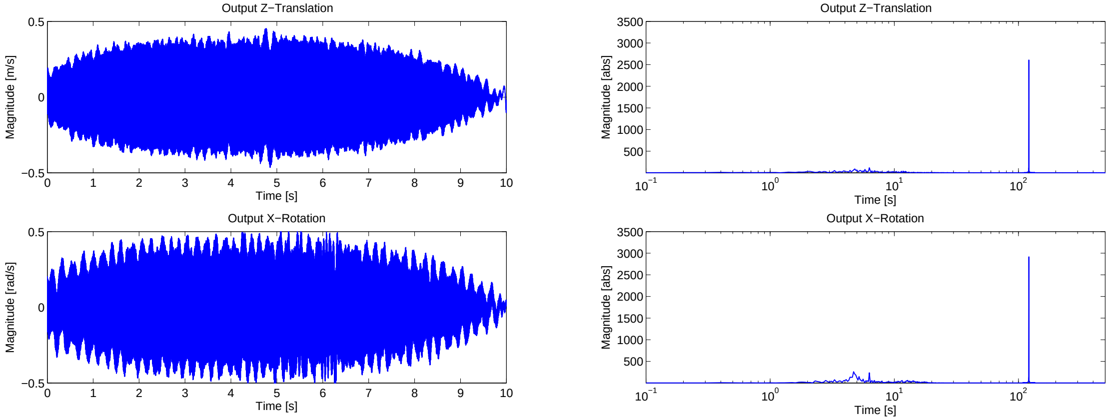
Data-driven estimation: experimental case study

- Output after 2 iterations: dominant frequency components

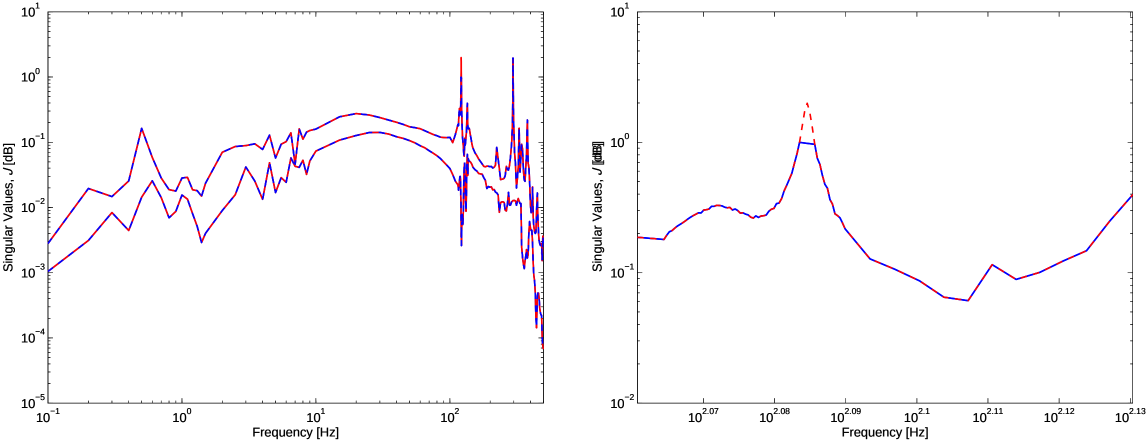


Data-driven estimation: experimental case study

- Output after 40 iterations: single component at $\omega^* = 121$ Hz
- $\omega^*_{FRF} \neq \omega^*_{\text{data-driven}}$, (121 Hz $\neq$ 296 Hz)



Data-driven estimation: experimental case study



- initial long measurement: baseline, different frequency?
- much longer validation measurement: we missed a peak!
- iterative approach: useful for experiment design

Overview

Conclusion

- unified overview of data-driven adjoint estimation algorithms

Extensions: a unified and broad framework!

- multivariable (Oomen et al.2014) (Bolder et al.2018) (Aarnoudse & Oomen2021)
- reset vs. continous operation (Oomen & Rojas2024)
- noise analysis (Rojas et al.2012)
- connections to VRFT (Formentin et al.2015) IFT (Aarnoudse et al.2025)
- many many extensions, other properties, etc.: paper in preparation

Aarnoudse, L., den Toom, P. & Oomen, T. (2025), ‘Randomized iterative feedback tuning for fast MIMO feedback design of a mechatronic system’, *Control Engineering Practice* **154**, 106152.

Aarnoudse, L. & Oomen, T. (2021), ‘Model-free learning for massive MIMO systems: Stochastic approximation adjoint iterative learning control’, *IEEE Control Systems Letters (L-CSS)* **5**(6), 1946–1951.

Bolder, J., Kleinendorst, S. & Oomen, T. (2018), ‘Data-driven multivariable ILC: Enhanced performance by eliminating L and Q filters’, *International Journal of Robust and Nonlinear Control* **28**(12), 3728–3751.

Dinkla, R., Oomen, T., Mulders, S. P. & van Wingerden, J.-W. (2024), Data-enabled predictive repetitive control, *in* ‘Proceedings of the 63rd Conference on Decision and Control’, Milan, Italy, pp. 6749–6754.

Formentin, S., Bisoffi, A. & Oomen, T. (2015), Asymptotically exact direct data-driven multivariable controller tuning, *in* ‘ 2015 IFAC Symposium on System Identification’, Beijing, China, pp. 1349–1354.

Jiang, Z. & Chu, B. (2022), ‘Norm optimal iterative learning control: A data-driven approach’, *IFAC PapersOnLine* **55**(12), 482–487.

Oomen, T. & Rojas, C. R. (2024), ‘Reset-free data-driven gain estimation: Power iteration using reversed-circulant matrices’, *Automatica* **161**, 111505.

Oomen, T., van der Maas, R., Rojas, C. R. & Hjalmarsson, H. (2014), ‘Iterative data-driven $\mathcal{H}_\infty$ norm estimation of multivariable systems with application to robust active vibration isolation’, *IEEE Transactions on Control Systems Technology* **22**(6), 2247–2260.

Rojas, C. R., Oomen, T., Hjalmarsson, H. & Wahlberg, B. (2012), ‘Analyzing iterations in identification with application to nonparametric $\mathcal{H}_\infty$-norm estimation’, *Automatica* **48**(11), 2776–2790.